TECHNICAL AUDIT + PRACTICAL GUIDE

ParkinSense PK

Project ka complete workflow, software checks aur real-recording evidence

Updated 19 October 2026 | Roman Urdu + English technical terms

Software checks: PASS

463 Python tests passed; zero failures, zero skips. Android build, 12 unit tests aur lint checks successful. Real media par movement, hand aur voice pipelines run ki gayi hain.

Clinical validation: abhi mukammal nahi

Recording accept hona diagnosis, measurement accuracy ya task ki tasdeeq nahi. Is audit mein real task-mismatch inputs bhi quality gate pass kar gaye. Har module ki actual coverage aur baqi kaam andar clearly diya gaya hai.

Aap ke liye

Ye document samjhata hai ke recording se measurements, 3D view, personal history aur report tak data kaise guzarta hai. Is mein code ka component map, exact test evidence, practical istemal aur product ko clinic pilot tak le jane ke aglay steps shamil hain.

Audit scope

90 first-party Python source files ka inventory; web, backend, ML utilities, Java Android app, JavaScript renderer, storage, infrastructure aur tests ka review. Manual review critical workflows par focused tha; test count code-coverage percentage nahi hai.

Real patient account ki saved history ko is audit mein modify nahi kiya gaya. Public research media aur isolated test databases istemal huay. Ye technical project report hai, kisi patient ki medical diagnosis report nahi.

01 / Report ko kaise parhein

Hissa	Aap ko kya milega
Audit result & workflow	Software evidence, architecture aur six patient sections.
Movement modules	Walking, turning, dono haath, sit-to-stand, reach, grasp aur object transfer.
Voice & daily life	WAV processing, reviewed text, ADL aur human-authored care notes.
3D & reporting	Live overlays, world XYZ replay, statistical graphs, baseline comparison aur patient report.
Operations	Accounts, roles, private media, consent, backups, Android, validation aur clinic pilot.
Real evidence	Published recordings, outputs, deliberate wrong-task checks aur missing clinical evidence.
Appendices	Test suites, first-party source map, configuration and source references.

Teen alag meanings

• PASS: automated software assertion ya build check successful hua.

• ACCEPT: local recording-quality rules ko usable input mila. Is se task identity ya clinical correctness prove nahi hoti.

• VALIDATED: independent reference measurements, representative participants aur predeclared protocol ki zarurat hoti hai. Is project ke liye yeh status abhi establish nahi hua.

Is audit mein naya kya fix hua?

Purani incompatible visit baseline search ko rok rahi thi. Ab report coordinator aur web review matching method, task aur language wali pehli eligible visit tak search jari rakhte hain. Is ke liye six parameterized regression cases add huay. Accepted movement/voice review par English, Urdu aur Roman Urdu mein explicit task-review guidance bhi add hui.

Code: ml/temporal/baseline.py; app/views.py; app/components/patient_language.py; tests/test_intelligence.py; tests/test_baseline_selection.py

02 / Verified status aur boundaries

Check	Observed result	Is ka matlab
Python final suite	463 pass / 0 fail / 0 skip	Full collected suite; real model-loading mode aur isolated PostgreSQL security tests enabled.
Android	Build successful; 12 unit tests pass	Debug preview builds. Intended physical phone ka test abhi nahi hua.
Lint & style	Ruff pass; diff check pass	Fresh offline Android lint: 0 errors, 23 warnings (earlier cached report had 33). Warnings release readiness ka remaining work hain.
Real video	11 runs; 4 matched demo analyses accepted	4 matched analyses = 2 gait excerpts + 2 hand-side analyses. Ye 4 independent patients nahi.
Real audio	PD-labelled source + excerpt; ordinary narration	Usable acoustic signals measured. Recording content/task completion human-confirmed nahi.
Reports & 3D	11 video HTML reports; 4 matched 3D replays	Local generation verified. Historical browser report-download completion not verified.

Warnings

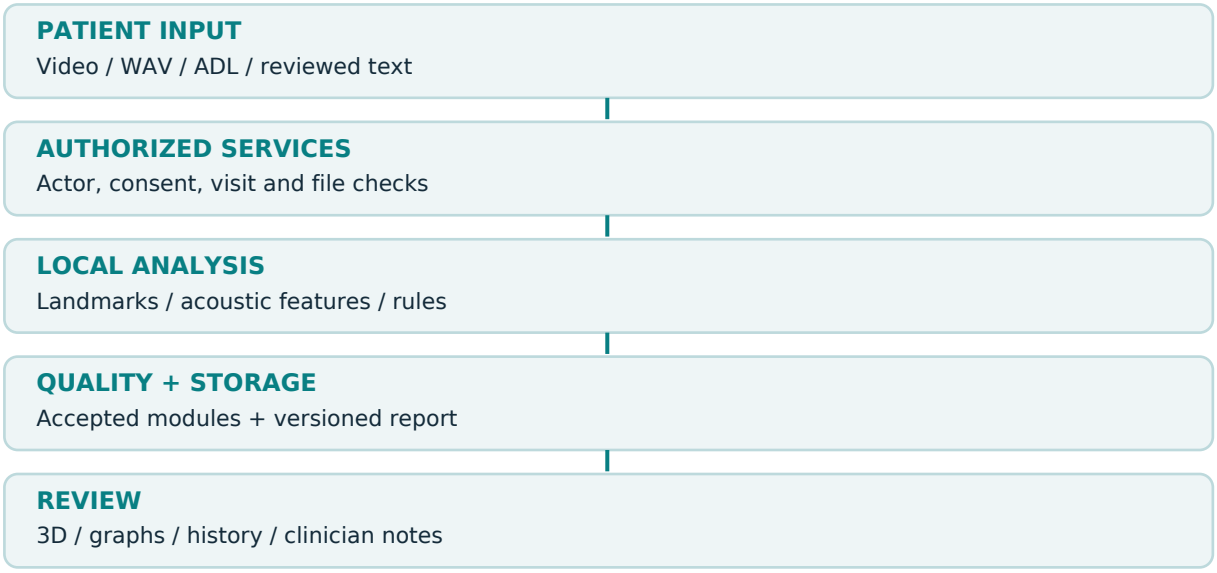
Python suite mein Starlette/httpx test-client deprecation warning hai. Android lint ke 23 warnings mein 18 text/localisation, 9 dependency updates aur baqi preview-network, lifecycle/orientation aur persistence items hain. Zero lint errors ka matlab zero product risk nahi.

Coverage jo available nahi

Matched clinical recordings for sit-to-stand, functional reach, grasp, object transfer aur hand opening/closing collect nahi huin. Dedicated turning aur sustained-vowel validation bhi nahi hui. Real phone hardware, independent clinical ground truth, multiple-camera cohort aur fresh repeated acquisitions is run mein available nahi the.

Final suite runtime: 179.633 seconds. Source state is local working tree, audit 18 October and update 19 October 2026; no release or deployment certification.

03 / Camera se report tak architecture



Streamlit app user interface aur navigation handle karti hai. UI ka session active visit ya temporary draft ko yaad rakhta hai; saved medical records ka authoritative store PostgreSQL hai. Files private filesystem mein, database references ke saath rehti hain.

Web, mobile API aur optional MCP facade ek hi authorized service layer par depend karte hain. Har request mein ownership/clinician linkage aur consent check hota hai; sirf screen chhupa dena security boundary nahi.

AI kahan hai?

MediaPipe trained models body aur hand landmarks estimate karte hain. Features, quality checks, acoustic signal processing, keyword matching, graphs aur report wording largely deterministic hain. Clinical severity model, learned fusion, automatic speech transcription aur external LLM production path mein configured nahi.

Code: app/Home.py; app/views.py; backend/services/; backend/mobile_api.py; ml/vision/pipeline.py; ml/inference/reports.py

04 / Patient ke six assessment sections

Section	Selectable tasks	Input/output
Walking	gait, turning	Body video; stepping/orientation proxies.
Hand	finger_left, finger_right, hand_open_close	Selected hand video; aperture/opening cycles.
Sit-to-stand	sit_to_stand	Body video; knee-extension transition proxy.
Reach / object tasks	reach, grasp, object_transfer	Body or hand video; relative reach/grip movement.
Voice & description	vowel, sentence, natural; reviewed text	WAV acoustic measures plus separately reviewed personal text.
Daily activities	9 ADL responses + optional note	Explicit 0-4 self-reported difficulty; no video inference.

Aik visit ka practical flow

Login karein, Start or resume an assessment kholein aur existing draft select karein. Section kholne se nayi duplicate visit nahi banti. Task choose karein, us ki local instructions parhein, recording upload ya live capture karein, analysis ke baad result review karke save karein.

Har section isi visit ke andar apna module save karta hai. Save accepted results ke liye enabled hota hai. Retry result par reason aur step-by-step guidance milti hai. Sab desired tasks ke baad visit submit/review karein; clinician notes aur personal report usi visit ke context mein rehte hain.

What was tested

Isolated login se all six sections khole gaye, task selectors change huay aur same draft preserved raha. ADL aur reviewed text resume tests ne saved answers, failed edits aur separate visits ki draft isolation verify ki.

Code: tests/test_assessment_sections.py; tests/test_voice_adl_resume.py; app/components/patient_language.py

05 / Walking aur turning ka analysis

Walking / gait

Pose detector 33 body landmarks estimate karta hai. Required shoulders, hips, knees, ankles aur feet ki visibility check hoti hai. Image coordinates aspect-ratio corrected hotay hain; relative distances trunk scale se normalize hoti hain. Knee angles ke liye estimated world coordinates use hotay hain jab available hon.

Ankle separation ki time series se cycles aur cadence proxy banta hai. Knee range, hip vertical range, ankle asymmetry aur reliable wrists milne par arm-swing proxy report hota hai. Missing arm tracking ko zero arm movement nahi kaha jata; metric unavailable rehta hai.

Turning

Shoulder orientation ka change aur active turn-duration proxy nikala jata hai. Signal ko continuous tracking segments mein process kiya jata hai taake missing-frame gaps par artificial angular jumps ya transitions count na hon.

Real input	Observed output
Published PD outdoor excerpt	49 / 60 sampled frames usable (81.7%); 2 gait cycles; cadence proxy 49.95 steps/min.
Published PD indoor excerpt	60 / 60 usable (100%); 3 cycles; cadence proxy 60.96 steps/min; arm-swing unavailable.
Walking selected as turning	Accepted by consistency rules. This is not evidence that the instructed turning protocol was completed.

Interpretation

81.7% aur 100% tracking coverage hain, clinical accuracy nahi. Published case ke dono clips ek patient ke hain. Camera motion, viewpoint aur short duration cadence/range ko affect kar sakte hain; reference counting independently verify nahi hui.

Code: ml/vision/tasks.py; ml/vision/pipeline.py; ml/vision/features.py; tests/test_gait_feature_visibility.py; tests/test_movement_task_features.py

06 / Hand, finger tapping aur grip

Landmarks se measures

Hand Landmarker 21 joints detect karta hai. Left/right tasks anatomical side ko filter karte hain; initial hand side lock aur wrist continuity selection hand-switching reduce karte hain. Unmirrored input aur selected side ka human confirmation zaroori hai.

Thumb-index distance ko wrist-to-middle-MCP hand scale se divide kiya jata hai. Tapping mein cycles, cycle frequency, range, inter-cycle variation aur late/early range ratio derive hotay hain. Hand opening mein fingertips-to-wrist opening signal use hota hai. Wrist-path proxy image-plane motion hai, measured room trajectory nahi.

Published hand recording	Left analysis	Right analysis
Usable sampled frames	177 / 177	177 / 177
Detected cycles (proxy)	21	21
Frequency (proxy)	1.79 Hz	1.79 Hz
Input identity	Same research video	Same research video

Evidence ki had

Figshare source finger-tapping demonstration hai; is individual ka Parkinson diagnosis source metadata se establish nahi hua. Dono sides ko do independent participants count nahi karna. Exact cycle count ko blinded manual reference ke against validate nahi kiya gaya.

Grasp aur transfer

In tasks mein available measurement grip aperture aur hand movement hai. Object detector, contact sensor ya successful transfer verification configured nahi. Isi liye finger-tapping video grasp/transfer ke taur par bhi quality gate pass kar gayi; clinical task completion manual review maangti hai.

Code: ml/vision/features.py; ml/vision/pipeline.py; app/components/hand_3d_viewer.py; app/components/pose_3d_viewer.py

07 / Sit-to-stand, reach aur task identity

Sit-to-stand

Left/right knee angles ka mean smooth signal banta hai. Low-to-high knee extension transition ka elapsed time aur angular range proxy report hota hai. Timestamp gaps bridge nahi kiye jatay. Ye feature engineering hai; validated sitting/standing action recognizer nahi.

Functional reach

Shoulder-to-wrist distance ka range trunk lengths mein express hota hai. Shoulders, elbows, wrists aur hips required hain; knees is upper-body task ke visibility gate ka hissa nahi. Target distance, grasp success ya fall risk infer nahi hota.

Stress input	Observed result	Meaning
Walking as sit-to-stand	ACCEPT; extension proxy returned	Knee motion alone cannot prove a chair-rise task.
Walking as reach	REPEAT; 17/60 usable frames	Arm visibility insufficient. Rejection is not proof of task classification.
Finger tapping as open/close	ACCEPT	Opening motion proxy can respond to another hand action.
Finger tapping as grasp/transfer	ACCEPT for both	Object interaction is not checked.

Priority before unsupervised use

Task ki visible review ko workflow ka essential hissa rakhein. Matching labelled recordings aur independent annotation collect karke task-specific eligibility criteria evaluate karein. Sirf is small sample ko reject karne ke liye arbitrary thresholds tune karna reliable fix nahi hoga.

In modules ke deterministic feature tests pass hain aur real inputs par code execute hua. Matching patient-task validation ko pending mark kiya gaya hai.

Code: ml/vision/features.py; ml/vision/tasks.py; tests/test_movement_task_features.py; app/views.py

08 / Voice recording aur acoustic analysis

Input contract

Local WAV input mono/stereo, 8-192 kHz aur maximum 180 seconds accept hota hai; processing file cap 72 MB hai. Vowel ke liye minimum 2 seconds, sentence/natural ke liye 3 seconds. Silent, clipped, invalid ya too-short samples repeat result dete hain.

Processing pipeline

Original file preserve hoti hai. Working signal se DC offset remove, polyphase resampling se 16 kHz aur spectral working copy ki peak normalization hoti hai. 25 ms windows aur 10 ms hop par energy activity; 0.2-second ya lambi interior silences pause statistics banati hain. Spectral centroid, 13 MFCC means aur sufficiently periodic signal par autocorrelation pitch estimate nikalta hai.

MDVR-KCL PD-labelled reading excerpt	Measured value
Prepared excerpt	Original recording ke 30-60 seconds; 30.00 s total
Source sampling	44.1 kHz, mono; exported PCM16
Signal-active fraction	72.48%
Internal pauses	6; total pause time 4.54 seconds
Estimated pitch median	120.30 Hz
Clipped samples	0.00%

Complete source 157.56 seconds ka tha aur woh bhi accepted hua. Excerpt ke numbers sirf is excerpt ke hain. PD label research dataset se aaya hai, app ne disease detect nahi ki. Speaker/task content ka independent listening annotation is audit mein nahi hua.

Task confirmation

Ordinary book narration vowel, sentence aur natural teeno selections par signal-quality checks pass kar gayi. System acoustic signal measure karta hai; sustained vowel ya prescribed sentence ki completion verify nahi karta. ASR, speech rate, jitter aur shimmer available nahi.

Code: ml/audio/features.py; app/views.py; tests/test_intelligence.py; tests/test_voice_adl_resume.py

09 / ADL aur reviewed personal description

Daily activities

Dressing, feeding, grooming, bathing, writing, phone use, walking, transfers aur household activities ke nine explicit responses required hain. Scale 0 = no difficulty se 4 = unable without help tak hai. Blank answer ko zero nahi banaya jata; boolean/invalid values reject hoti hain.

Result har response aur arithmetic mean retain karta hai. Optional other difficulty note 2,000 characters tak hai. Ye custom unvalidated questionnaire hai, UPDRS ya disease-severity scale nahi. Video se ADL answers automatically fill nahi hotay.

Reviewed text / NLP

Patient apni difficulty apne alfaz mein likhta hai ya reviewed transcript enter karta hai. Local multilingual dictionaries aur negation rules functional domains ke matched difficulty phrases identify karte hain. Short excerpts aur original reviewed text context ke liye preserve hotay hain.

Rule match 1 ka matlab phrase mila; 0 ka matlab matching phrase nahi mila. 0 ko confirmed normal function nahi kehna. Falls/freezing jaisay self-reported concerns review flags ban sakte hain. No automatic diagnosis, no freeform LLM-generated medical advice.

Human touch

Personal notes, daily check-ins, patient goals aur clinician-authored follow-up measured proxies se alag rehte hain. Is se user ki rozmarrah life aur priorities visible rehti hain, bina unhein fake AI score mein convert kiye.

Resume behavior

Saved answers aur notes same draft mein wapas khulte hain. Failed edit saved report ko overwrite nahi karta. Unsaved prose doosri visit mein copy nahi hoti. In behaviors ke UI regression tests pass huay.

Code: ml/tabular/adl.py; ml/nlp/functional.py; backend/services/care.py; tests/test_voice_adl_resume.py; tests/test_care.py

10 / Live camera aur professional 3D

Camera overlay aur 3D view

Camera image par lines detected joints ki projected screen position follow karti hain; depth styling genuine model world-depth se aati hai. Saath wala rotatable viewer world X/Y/Z use karta hai. Camera video khud 2D image hai; adjacent model estimated 3D articulation dikhata hai.

Body ka origin hips ke paas aur hand ka origin hand center ke paas hota hai. Scale MediaPipe-estimated meters hai. Global walking route, measured floor aur calibrated clinical distances available nahi. Solid avatar observed joints ke around display mesh hai, scanned skin/body surface nahi.

Live reliability

Asynchronous camera processor ko UI polling resolve karti hai. Stale observations expire hotay hain; lost tracking par fabricated motion draw nahi hoti. Short trails bounded hain. Persistent Plotly WebGL host latest figure update karta hai, drag/rotate ke dauran unnecessary remount avoid karta hai.

Saved replay

Play/pause, time slider, playback speed, camera presets, solid/skeleton styling aur joint trails available hain. Optional visual smoothing stored landmarks ya numerical measures ko change nahi karti. Compatible baseline gold overlay local origin aur relative recording time par align hoti hai.

Verification boundary

World-coordinate geometry, visibility masks, stale-state behavior, event ordering aur renderer update lifecycle automated tests se checked hain. Prior public hand replay ka browser playback observed hua. Is run mein physical camera capture/permissions aur all devices par live FPS benchmark nahi hua.

Code: ml/vision/spatial.py; ml/vision/live_overlay.py; app/components/live_camera.py; live_3d_viewer.py; live_plot.py / live_plot.js; pose_3d_viewer.py; motion_studio.py

11 / 3D statistics aur visit comparison

Graphs ki axes

Movement curves mein X = recording time, Y = measurement series aur Z = actual proxy value hota hai. Measurement summary mein values apni defined unit mein dikhte hain. Visit history accepted compatible dots dikhati hai. Voice, text aur ADL ko unke measured/self-reported values se 3D statistical view milta hai; un se fake body model nahi banta.

Different units ko aik value axis par mix nahi kiya jata. Missing observations blank rehti hain; missing values ko zero ya artificial normal score nahi banaya jata. Curves maximum 300 actual display samples plus gap separators tak bounded hain; descriptive mean/range/population SD full finite extracted samples se bante hain.

Baseline logic

Reference current visit se genuinely earlier, accepted aur matching task/method/language ka hona chahiye. Change = current minus reference. Percentage change sirf non-zero reference par meaningful formula se nikalta hai. Ye arithmetic change hai; clinical improvement ya deterioration ka validated threshold nahi.

Is audit ka fix

Pehle agar sab se purani accepted result ka method/language/task mismatch hota to search matching later reference tak nahi jati thi. Ab incompatible entries skip hoti hain. Aisi history jahan koi compatible reference nahi, wahan comparison unavailable rehta hai.

Graph ko kaise samjhein

Camera distance, light, task duration aur technique different hon to change recording setup ki wajah se bhi ho sakta hai. 3D presentation measurement ko clinically accurate nahi banati. Exact values aur source/unit labels graph ke saath dekhein.

Code: ml/common/statistics.py; app/components/statistical_3d.py; ml/temporal/baseline.py; app/views.py; tests/test_statistical_3d.py; tests/test_baseline_selection.py

12 / Automatic report aur fault guidance

Report kab banti hai?

Accepted module save hone par service same database transaction mein report content refresh karti hai. Source state ka stamp duplicate report creation avoid karta hai. Report mein task quality, actual measurements/units, source, eligible reference differences, limitations aur self-reported context shamil hotay hain.

Rejected attempt

Poor recording save gate ko pass nahi karti. Attempt preview phir bhi issue aur next steps explain kar sakta hai. Missing model, corrupt media, lighting, visibility, timing gaps aur quiet/clipped audio ki fixed catalogue guidance hai. Raw exception/private path ko patient-facing error mein expose nahi kiya jata.

Patient aur clinician wording

Fixed local templates structured results ko readable summaries mein badalti hain. Patient ke liye simpler wording aur clinician ke liye traceable measurements/review items hotay hain. System medication changes ya unsupported diagnosis generate nahi karta. Urdu/Roman Urdu product copy bhi human review maangti hai.

Export

App patient report print-ready HTML generate karti hai; browser Print se PDF ban sakti hai. Private reports/replays protected short-lived links se milte hain. Is audit ka detailed PDF alag standalone technical document hai. Public demo outputs mein source attribution attach hoti hai.

Verification

Report generation, escaping, provenance, invalid input, idempotency, stale revisions aur module-save/report atomicity tests pass huay. Browser report download ke pehle interrupted check ko completed claim nahi kiya gaya; local generation independently verified hai.

Code: backend/services/reporting.py; ml/inference/patient_report.py; ml/inference/reports.py; ml/common/guidance.py; app/components/analysis_feedback.py; private_access.py

13 / Accounts, roles aur persistence

Layer	Implemented behavior
Patient	Own visits, uploads, reports, consent and personal care records.
Clinician	Only authorized/linked patient records; authored review and pilot/study ownership.
Administrator	Staff/link management, recovery and request handling; not blanket patient-video access.
Database	PostgreSQL runtime; SQLAlchemy transactions; SQLite only explicitly enabled for tests/development.
Passwords	Salted scrypt hash; bounded input; constant-time digest comparison.
Files	UUID storage keys, path containment, private directory/file permissions and byte limits.

Security checks server par

Actor identity authenticated server session se aati hai. A client-supplied patient ID, file ID ya assessment ID ownership permission replace nahi karta. Draft status, current consent aur active account validation costly processing se pehle hoti hai. Failed upload/write paths cleanup aur rollback handle karte hain.

Private media access

Media gateway short-lived capability token issue karta hai, default expiry 180 seconds. Database mein token hash aur scope rehta hai. Authorization mein actor session revision, consent, linkage aur revocation check hotay hain. Private recordings Streamlit ke public media store mein nahi bheji jati.

Concurrency

Isolated PostgreSQL tests ne grant/recovery race paths exercise kiye. Actor/patient row locking aur single-use recovery semantics stale access ko prevent karne ke liye hain. Ye application-level security evidence hai; external penetration test ya deployment certification nahi.

Code: backend/services/__init__.py; backend/production.py; backend/security/; backend/storage/; backend/db/; scripts/secure_media_server.py; tests/test_production_security.py

14 / Consent, recovery aur backups

Consent & data handling

Monitoring consent withdraw karne par new monitoring writes aur issued media grants restrict/revoke hotay hain. Existing saved records automatically delete nahi hotay. Patient access, export ya erasure request create kar sakta hai; administrator us ka handling status update karta hai.

Important product boundary

Data-request status ko resolved set karna actual erasure/export operation ka proof nahi. Retention duration, responsible operator aur documented fulfillment process deployment se pehle define karna baqi hai.

Account recovery

Clinic/operator identity verify karne ke baad single-use recovery code issue karta hai. Expiry, attempts aur session revision protection use hoti hai; reset ke baad old sessions/grants invalid ho sakte hain. Automated email delivery ya identity verification provider configured nahi.

Encrypted backup

Backup script database aur private media snapshot banati hai. Separate passphrase se Scrypt-derived key aur AES-GCM encryption use hoti hai. Writes paused hone ki operator attestation required hai; backup script production maintenance ko khud orchestrate nahi karti. Changed media/checksum errors fail safely.

Restore drill

Restore archive authentication, safe paths, checksum aur schema verify karta hai; existing destination overwrite nahi karta. Current drill nayi isolated SQLite/media directory mein restore karta hai. Live PostgreSQL production recovery, offsite scheduling, retention aur key escrow operationally rehearse karna abhi baqi hai.

Automated tests isolated temporary data par chale; real patient database ka backup/restore ya erasure is audit mein execute nahi hua.

Code: backend/services/privacy.py; backend/production.py; scripts/account_recovery.py; backup_private.py; restore_drill.py; tests/test_privacy_integration.py; tests/test_production_security.py

15 / Android patient app

Native preview

Java Android app browser wrapper nahi. MainActivity login, patient dashboard/history/reports aur forms manage karti hai. LiveCheckerActivity CameraX video aur AudioRecord microphone capture karti hai. Server-side analysis same Python services use karti hai.

On phone	On server
Video brightness, edge/contrast and pixel change	MediaPipe landmarks, task features and accepted-result storage.
Mono 16 kHz microphone WAV; RMS/clipping/silence	Acoustic processing and report creation.
Review, discard and explicit upload controls	Ownership, consent, task, file and result validation.
Encrypted token in Android Keystore	Expiring 12-hour signed sessions; hashed nonce registry and reset revision.

Recording maximum 45 seconds; movement capture also 30 MB bounded. Imported files have a 100 MB cap. Backgrounding/leaving recorder releases hardware and discards unused capture. Server change logs out and invalidates pending callbacks; API client does not follow redirects.

Local vs release

Development preview allows private Wi-Fi HTTP, but rejects public HTTP origins. Release network configuration requires HTTPS. Laptop API on port 8767 must be reachable; 127.0.0.1 on the phone means the phone itself. API server ke liye current laptop LAN address use karein, purana hard-coded address assume na karein.

Evidence

Debug build and v2 signature verified. 12 unit tests rerun successfully (7 signal-math, 5 server-origin cases). Offline lint rerun: 0 errors, 23 warnings; this does not verify current dependency versions. Physical phone/emulator UI, denied camera/mic permissions aur hardware compatibility manual tests abhi pending. Android live preview currently gives quality signals; detailed articulated 3D review is the web workflow.

Code: mobile/android/app/src/main/java/org/parkinsense/pk/; backend/mobile_api.py; backend/mobile_auth.py; docs/mobile-android.md

16 / Real-world validation framework

Implementation

Clinician ek exact task, metric, unit, method version aur reference protocol define karta hai. Patient participation separate consent se hoti hai. Source measurement, hash aur assessment revision freeze hotay hain; later data changes ko purani evaluation mein silently merge nahi karna chahiye.

What the evaluator calculates

Held-out evaluation rows par participant-balanced bias, MAE aur RMSE bante hain. Prolific participant ko dozens of recordings ki wajah se extra weight nahi milta. Three ya zyada paired participants par deterministic bootstrap MAE interval available hota hai; tiny data par missing uncertainty ko zero nahi banaya jata.

Repeatability sirf explicitly matched repeat groups par niklti hai, same participant/device/lighting/site context ke saath. Within-participant SD aur approximate repeatability coefficient report hotay hain. True patient change ko measurement noise maan lena valid nahi.

Abhi actual evidence kya hai?

Evaluator ki math, authorization, consent, duplicates aur protocol behavior automated tests se checked hain. Current public demos ke liye blinded reference measurements ya repeat acquisitions nahi mile. Is liye clinical MAE, sensitivity, specificity, agreement aur camera-to-camera repeatability numbers report nahi kiye gaye.

A defensible next study

Clinician ke saath protocol aur acceptable error pehle define karein; representative people, cameras, lighting, body/hand visibility aur languages plan karein. Har attempt log karein, rejected clips ko denominator mein rakhein. Independent blinded annotation aur participant-separated evaluation use karein. IMDRF guidance representative data aur independent testing ko central rakhti hai [S5].

Code: backend/services/validation.py; ml/validation/evaluation.py; app/components/validation_review.py; scripts/evaluate_recordings.py; tests/test_measurement_validation.py

17 / Clinic pilot aur care workflow

Feasibility, not invented validation

Clinician pilot dates aur targets define karta hai, patients invite karta hai aur separate participation acknowledgement record hoti hai. First outcome/timer ke baad plan freeze hota hai. Withdrawn ya no-longer-authorized patients aggregate se exclude hotay hain.

Pilot outcome	How it is tracked
Review time	Paired baseline/app durations; measured timers or explicitly sourced observations; median saving.
Task completion	Planned, attempted, completed; unattempted work denominator mein retained.
Recording usability	Usable / attempted recordings; minimum participant coverage.
Patient usability	1-5 rating, need-for-help, language/device context.
Commercial intent	Clinic-owner/procurement/clinician stated willingness, currency and period; not actual payment.

Default feasibility targets: minimum 10 patients, 10 review pairs, 10 usability responses, median review-time saving 20%, task completion 85%, usable recordings 85%, usability rating 4/5. Ye configurable project targets hain; international clinical approval thresholds nahi.

Human-authored care

Energy, sleep quality, movement ease aur mood ke daily 0-4 ratings wellbeing/self-report hain. Falls can be yes/no/unknown. Patient apne personal goals likhta hai; clinician follow-up plan aur next review date author karta hai. Source aur author visible rehte hain.

What remains

Software mein pilot management available hai, lekin actual clinic enrollment, measured time savings, patient completion rates aur willingness-to-pay evidence is audit ne generate nahi kiya.

Code: backend/services/pilot.py; backend/services/care.py; app/components/clinic_pilot.py; care_panels.py; tests/test_clinic_pilot.py; tests/test_care.py

18 / Training, explainability aur optional MCP

Developer training utilities

Feature export quality-gated numeric results ko aligned table mein la sakta hai. Patient identifiers pseudonymous hain, automatically anonymous nahi. Baseline training utilities patient-disjoint splits, training-only preprocessing aur provenance manifest use karti hain. In utilities ko test karna clinical model train/approve karne ke barabar nahi.

Experimental inference

Trusted local artifact directory, manifest purpose/schema, feature list aur SHA-256 verify karke experimental baseline load ho sakta hai. Joblib artifacts executable code carry kar sakte hain, is liye arbitrary uploaded models ka loading interface nahi. Patient-facing validated prediction model currently unavailable hai.

Fusion & explanations

Fusion four modalities - movement, audio, text aur ADL - ki data availability aur actual feature coverage summarize karti hai. Coverage percentage disease score nahi. Explainability descriptive reference differences batati hai; calibrated individual uncertainty, SHAP-based clinical explanation ya learned attention model configured nahi.

Evidence retrieval

ApprovedKnowledgeBase opt-in versioned approved documents retrieve kar sakta hai. Default empty hai; internet se automatic unreviewed medical advice inject nahi hoti. Report external LLM ko raw patient recordings ya prose transmit nahi karti.

Optional MCP facade

Typed localhost tools server-configured actor se bound hain. Client actor, SQL ya arbitrary file paths choose nahi kar sakta. Structured outputs raw media/prose aur internal storage details omit karte hain. HTTP serving is optional, default application ka autonomous agent workflow nahi.

Code: ml/tabular/training.py; ml/inference/models.py; evidence.py; ml/multimodal/fusion.py; ml/xai/explain.py; backend/mcp_tools.py; scripts/train_baseline.py

19 / Real-media evidence: exact scope

Source / selected input	Tested	Result and caveat
Schaeffer et al. 2016 PD case [S1]	2 six-second gait excerpts	Both accepted. Same patient, different excerpts; diagnosis from publication, not inferred.
Kasemsap 2025 finger video [S2]	11.8 s recording, left and right	Both accepted, 177/177 frames each. Participant diagnosis unspecified.
MDVR-KCL ReadText / PD [S3]	157.56 s complete WAV + fixed 30-60 s excerpt	Both accepted. PD label from dataset folder/metadata; not disease detection.
LibriSpeech/librosa narration [S4]	14.84 s speech; natural, sentence and vowel selections	All signal checks accepted. Not a Parkinson patient sample or sustained-vowel performance test.
Artificial dark derivative	Indoor gait brightness x 0.05	Rejected; 0/60 landmarks. A stress transformation, not an additional real environment/participant.

Why these inputs

Public licensed sources were chosen with provenance and bounded local preparation. Gait segments were visually selected before analysis; hand source kept at native timing. Voice excerpt fixed at 30-60 seconds before feature inspection. No repeated search for a clip merely to obtain green status was used.

Outputs retained

Audit folder contains per-video analysis.json, HTML report, matched 3D HTML replays, audio analysis JSON, MDVR extraction/preparation metadata and source hashes. These are local reproducibility artifacts; private patient visits were not populated with public demonstrations.

Evidence root: artifacts/project-audit-2026-10-18/real-media/. Media sources and CC BY 4.0 attribution are listed at the end.

20 / Tracking coverage is not accuracy

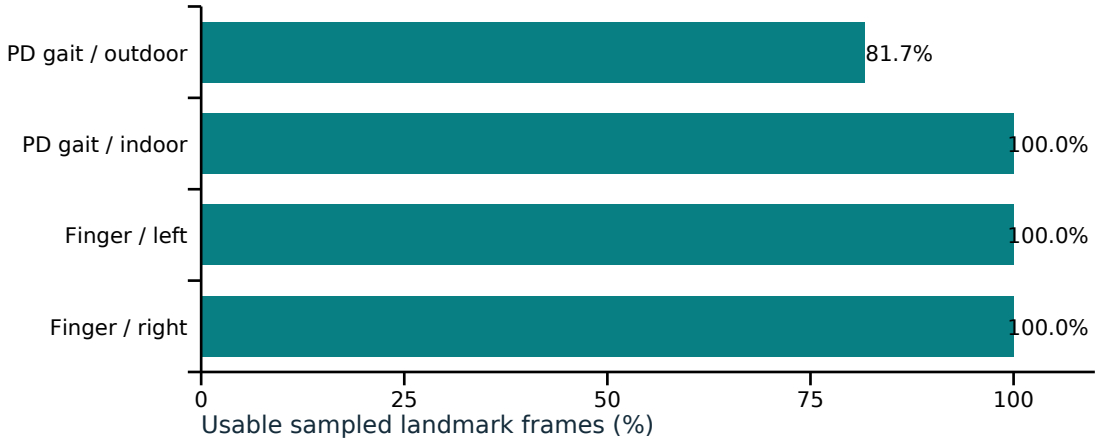


Figure: same four matched video analyses. Denominators: 60, 60, 177 and 177 sampled frames respectively. Full video-frame count is different from sampled-frame count. No clinical reference was used to calculate these percentages.

What these bars establish

Landmark detector ko in selected clips mein sufficient usable joints mile. World XYZ geometry available thi aur corresponding 3D replay HTML files generated huin. Indoor gait mein optional arm-swing metric absent rehna correct missing-data behavior tha.

What they do not establish

Bars se knee-angle accuracy, exact cadence, PD sensitivity, severity, diagnosis, calibrated distance ya patient-to-patient generalization conclude nahi ho sakti. Same file ko rerun karna camera/lighting repeatability experiment nahi.

Real-media observations are not an automated pass count

11 video processing runs mein 4 matched demos, 6 deliberate task-mismatch cases aur 1 dark derivative tha. 5/6 mismatch inputs accepted huay; reach rejected hua. Yeh limitation explicitly retained hai, green software suite ke andar hide nahi ki gayi.

21 / Remaining work, ranked by impact

Priority	Work	Completion evidence required
P1: task validity	Guided protocol + independent task annotation; object task verification or explicit reviewer workflow.	Matched and mismatched recordings per task; report acceptance and task-validity separately.
P1: measurement validity	Representative acquisition and blinded reference measurements.	Predeclared error limits; held-out participants; rejection coverage, MAE/CI and repeatability by setup.
P1: phone usability	Test intended Android phones and mobile browser capture; improve localisation and accessibility.	Permission denial, backgrounding, slow network, portrait/landscape, Urdu text and assisted task completion evidence.
P1: clinic pilot	Run a consented feasibility pilot with actual clinicians/patients.	Paired review times, completion/usability outcomes and separately documented payment intent.
P2: production operations	HTTPS deployment, recovery ownership, backup schedule, retention/erasure process and load testing.	Restore drill to target runtime, key handling, monitoring, independent security review and load evidence.
P2: maintenance	Resolve Android localisation/dependency warnings and test-client deprecation.	Compatible upgrade regression runs; clean review of remaining exceptions.

Is audit mein koi remaining automated test failure nahi hai. Upar wale items real-world evidence aur product readiness ki requirements hain; sirf extra unit tests se complete nahi hongay.

Readiness assessment

Supervised demonstration aur structured feasibility preparation ke liye useful working prototype. Unsupervised medical assessment, proven clinical accuracy ya production medical-device readiness ka claim abhi supported nahi.

22 / Aap project kaise chalayen

Existing local setup

Project directory mein terminal khol kar existing setup start karein. Docker/PostgreSQL aur installed Python dependencies required hain. Ye command database service, protected media gateway aur Streamlit UI start karti hai:

```
cd "/home/shaiby/Parkinsense PK"
bash scripts/run_local.sh
```

Browser: http://127.0.0.1:8501/ . Public demo bina login available hai. Apni private assessment ke liye apna account use karein. Missing landmark models ki error aaye to official downloader run karke dobara analyze karein:

```
uv run python scripts/download_landmark_models.py
```

Aik recording ka checklist

• Correct task aur anatomical side select karein; relevant joints poori clip mein visible hon.

• Stable camera, normal speed, even lighting aur same setup use karein. Video task minimum 3 ya 5 seconds; maximum 45 seconds / 100 MB.

• Voice WAV ke liye correct task, quiet environment aur reasonable microphone distance rakhein.

• Analyze karein; agar REPEAT ho to guidance follow karein. ACCEPT ke baad actual task match verify karke save karein.

• 3D ko rotate/play karke motion dekhein; units aur limitations ke saath values parhein. History/report mein genuinely matching previous visit compare karein.

Android API start

```
uv run python scripts/run_mobile_api.py
```

Phone testing ke liye trusted local network binding aur current laptop LAN address configure karein, jaisa docs/mobile-android.md mein diya hai. Production ke liye trusted HTTPS origin use hona chahiye. App account recovery code chat ya public report mein share na karein.

23 / Download and login follow-up

19 October 2026 update; read this with the earlier real-media evidence.

Login ke andar kya verify hua?

Disposable test account se actual login form use karke six assessment sections khole gaye. Same draft retention, har task selector, ADL/text resume, voice saves, saved report preservation aur ownership checks automated UI/service tests mein pass huay. Aap ke personal account ki har saved assessment ko individually verified claim nahi kiya gaya.

Download fix

Generated-at timestamp ki wajah se report bytes minute change par badal sakti thi. Is se unrelated UI rerun public download identity change karta tha; private attempt report ka prepared grant content stamp se match na kare to dobara preparation button aa sakta tha. Report HTML ab per-session source fingerprint ke saath stable rehta hai; data, actor, language ya selected example badlay to regenerate hota hai.

Public demo mein Open report here / Report yahin kholein button add hai. Ye wahi attributed HTML report app ke andar dikhata hai jo download control use karta hai. File format HTML hai; downloaded file ko browser mein khol kar Print / Save as PDF kiya ja sakta hai. Private patient downloads protected gateway par hi rehte hain.

New regression evidence

Advancing-clock tests verify that rerun se public report URL unchanged rehta hai, inline content exactly matches export, language change invalidates it, aur private attempt button prepared protected link tak pohanchta hai. Native browser download completion is update mein verify nahi hui: browser policy ne localhost page observation block ki. Is boundary ko code tests se browser success ke claim mein convert nahi kiya gaya.

Code: app/components/report_document.py; public_demo.py; analysis_feedback.py; tests/test_public_demo.py; tests/test_analysis_feedback_ui.py

23A / Test execution and reproducibility

Final run

Python 3.11 local virtual environment mein all collected tests run huay. PARKINSENSE_RUN_MODEL_TESTS=1 enabled tha. Security URL disposable PostgreSQL 16 container ke dedicated parkinsense_test_security_* database ko point karti thi; existing application database ko use nahi kiya gaya.

```
.venv/bin/python -m pytest -q -ra
.venv/bin/ruff check app backend ml scripts tests
git diff --check
```

Security integration ke liye isolated test database environment explicitly supply karna hota hai. Ordinary command without it PostgreSQL-only tests skip kar sakti hai; is audited final run mein zero skips the. Final helper records no password in report/log output and stops the temporary container.

Android

```
bash scripts/build_android_apk.sh
```

Build helper assembleDebug, testDebugUnitTest aur lintDebug run karta hai. Audit mein native unit task aur lint task explicitly rerun kiye gaye; compilation dependencies up-to-date ho sakti hain. Twelve unit tests ka latest XML current rerun ka hai.

Evidence files

File under audit folder	Purpose
pytest-final.log / pytest-final.xml	Final 463-test result and individual case evidence.
android-build.log / android-rerun.log	Debug build/signature and native checks.
ruff.log; component-inventory.json	Style result; source paths, definitions and SHA-256.
real-media/results.json	All 11 video runs, quality decisions, measurements and input hashes.
mdvr-source.json / mdvr-extraction.json	Source licence/metadata and bounded ZIP-member extraction.
run_final_suite.py / run_real_media.py	Local repeatable audit helpers; not application runtime modules.

24 / Automated test inventory

Test module	Cases	Status
test_3d_studio.py	15	PASS
test_analysis_feedback_ui.py	5	PASS
test_app.py	6	PASS
test_assessment_sections.py	6	PASS
test_auto_reports.py	11	PASS
test_backend.py	12	PASS
test_baseline_selection.py	3	PASS
test_care.py	28	PASS
test_care_panels.py	4	PASS
test_clinic_pilot.py	15	PASS
test_feature_tables.py	2	PASS
test_gait_feature_visibility.py	37	PASS
test_guidance.py	26	PASS
test_intelligence.py	22	PASS
test_live_3d.py	24	PASS
test_live_camera_ui.py	24	PASS
test_live_plot_runtime.py	1	PASS

All listed cases belong to the final run: 463 total, zero failures/errors/skips. Counts are not independent clinical participants or a code-coverage percentage.

24 / Automated test inventory (continued)

Test module	Cases	Status
test_live_tracking.py	17	PASS
test_measurement_validation.py	21	PASS
test_mobile_api.py	11	PASS
test_motion_studio_ui.py	9	PASS
test_movement_task_features.py	22	PASS
test_patient_usability.py	22	PASS
test_privacy_integration.py	7	PASS
test_production_security.py	40	PASS
test_public_demo.py	15	PASS
test_statistical_3d.py	31	PASS
test_vision.py	14	PASS
test_visit_summary.py	8	PASS
test_voice_adl_resume.py	5	PASS

All listed cases belong to the final run: 463 total, zero failures/errors/skips. Counts are not independent clinical participants or a code-coverage percentage.

25 / Sources, licences and preparation

S1

Eva Schaeffer, Gerd Linke and Daniela Berg (2016). "Cueing" for Levodopa-Induced Dyskinesias in Parkinson's Disease. Frontiers in Neurology. CC BY 4.0. Source publication documents PD case and publication consent.

Open original source

S2

Kasemsap (2025). Example Video Finger tapping for bradykinesia detection. Figshare, version 1. CC BY 4.0. Individual diagnosis not specified.

Open original source

S3

Hagen Jaeger, Dhaval Trivedi and Michael Stadtschnitzer. Mobile Device Voice Recordings at King's College London (MDVR-KCL). Zenodo record 2867216. CC BY 4.0; recordings made in 2017. Dataset ReadText/PD member used.

Open original source

S4

Librosa example recordings: LibriSpeech / Garth Comira narration (5703-47212-0000). CC BY 4.0. Natural speech demonstration; not a PD patient recording.

Open original source

S5

IMDRF. Good machine learning practice for medical device development: Guiding principles (2025). Used for evaluation-planning context, not as an endorsement of this project.

Open original source

CC BY 4.0 licence: https://creativecommons.org/licenses/by/4.0/ . Attribution accompanies public demos; adaptations are described below. Public availability is not evidence of clinical validation.

26 / Media provenance and audit limits

Video preparation

PD gait source OGV was transcoded to H.264 MP4, original 640 x 480 resolution and 29.97 fps. Outdoor excerpt 7.5-13.5 s; indoor excerpt 31.5-37.5 s. Audio removed; no spatial crop, enhancement, speed change or synthetic frames. Hand video retained full 11.8 s timing at 960 x 540 / 30 fps; source face blurring was already present.

Audio preparation

MDVR selected ZIP member was fetched with bounded HTTP ranges, decompressed and CRC checked; complete large dataset was not needed. Full WAV analyzed; fixed 30-60 s excerpt exported at original 44.1 kHz as PCM16, without pre-analysis denoising or normalization. Analyzer subsequently uses its own documented working-copy preprocessing.

```
MDVR source SHA-256:
32b7bc39988320bc490d9ef4084269ccb98c40a321bed887d9a24a19960e3be1
```

```
MDVR excerpt SHA-256:
30b93d5a0d7181269ae1ec9a605b5c2377b63bfb99383c3b6557fbf0d8e5bcfd
```

Reproducibility and judgement

Video input hashes are retained per result. Dimming x0.05 is labelled a synthetic stress derivative. No source performance claims were copied into the app. No manual reference cycle counts, blinded clinical ratings or exact-task labels were invented for these clips.

Scope limitations

Source inventory and software tests cover implementation, not every possible runtime state. There was no code-coverage measurement, load/penetration assessment, physical-phone acceptance test or representative clinical trial. Existing patient records were excluded. The original roadmap is planning context; implementation claims here come from inspected current code and observed evidence.

Appendix A / Python component map

Files 1-12 of 90. Paths relative to project root. All inventoried via AST; critical workflows manually reviewed.

Source file	Lines	Responsibility
app/Home.py	128	Single role-aware entry point; services enforce every data access as well.
app/__init__.py	1	ParkinSense-PK Streamlit application.
app/components/__init__.py	2	Reusable assessment review components.
app/components/analysis_feedback.py	70	Reusable analysis attempt feedback and printable patient report downloads.
app/components/care_panels.py	208	Patient words, personal goals and clinician-authored follow-up.
app/components/clinic_pilot.py	348	Human-run clinic pilots and separately acknowledged patient participation.
app/components/data_privacy.py	95	Plain-language data handling and patient control over monitoring consent.
app/components/design.py	289	Small, accessible visual primitives for the care workspace.
app/components/guidance_panel.py	45	Plain-language next steps, rendered from the fixed local guidance catalogue.
app/components/hand_3d_viewer.py	13	Hand articulation uses the same truthful model-local world coordinate studio.
app/components/live_3d_viewer.py	134	Current-frame focused 3D movement viewer for live recording.
app/components/live_camera.py	76	Polling controls that resolve the asynchronously created camera processor.

Exact definitions, line numbers and SHA-256 for each file are retained in component-inventory.json. Package __init__ files also appear so the source inventory is explicit.

Appendix A / Python component map

Files 13-24 of 90. Paths relative to project root. All inventoried via AST; critical workflows manually reviewed.

Source file	Lines	Responsibility
app/components/live_plot.py	42	Persistent WebGL host: update live data without remounting the plot.
app/components/motion_studio.py	94	Patient-friendly controls around the model-local 3D movement renderer.
app/components/patient_language.py	319	Locally written patient instructions, with stable task and response IDs.
app/components/patient_navigation.py	23	Patient navigation copy; callable page routes stay unchanged.
app/components/pose_3d_viewer.py	430	A bounded WebGL movement studio using genuine model-local world XYZ.
app/components/private_access.py	72	Private recordings/exports never enter Streamlit's public media storage.
app/components/public_demo.py	252	A public-video demonstration with no patient, database, or account access.
app/components/report_document.py	20	Stable report bytes for one browser session, invalidated by source changes.
app/components/statistical_3d.py	232	Professional 3D descriptive graphs for every implemented analysis module.
app/components/validation_review.py	139	Actual measurement studies, reference values and a native-unit 3D view.
app/components/visit_summary.py	83	Small read-only visit summaries without Streamlit's lazy DataFrame grid.
app/views.py	901	Patient, clinician and administrator experiences over authorized services.

Exact definitions, line numbers and SHA-256 for each file are retained in component-inventory.json. Package __init__ files also appear so the source inventory is explicit.

Appendix A / Python component map

Files 25-36 of 90. Paths relative to project root. All inventoried via AST; critical workflows manually reviewed.

Source file	Lines	Responsibility
backend/__init__.py	1	ParkinSense application services and private persistence.
backend/config.py	36	Runtime configuration. Private values are never returned to the UI.
backend/db/__init__.py	68	Database lifecycle and short transaction boundaries.
backend/db/models.py	204	Portable SQLAlchemy schema; PostgreSQL is the runtime persistence engine.
backend/db/pilot_models.py	84	Additive pilot records; workflow evidence is separate from clinical results.
backend/db/privacy_models.py	17	Patient-originated data requests without copying health information.
backend/db/production_models.py	61	Security state kept separate from clinical records and media bytes.
backend/db/validation_models.py	40	Additive, consented reference-measurement studies.
backend/mcp_tools.py	203	Optional typed MCP facade over ordinary authorized services.
backend/mobile_api.py	349	Patient-only native app gateway over the same authorized persistence services.
backend/mobile_auth.py	174	Expiring patient API sessions, tied to the existing password-reset revision.
backend/production.py	330	Revocable media grants and administrator-verified account recovery.

Exact definitions, line numbers and SHA-256 for each file are retained in component-inventory.json. Package __init__ files also appear so the source inventory is explicit.

Appendix A / Python component map

Files 37-48 of 90. Paths relative to project root. All inventoried via AST; critical workflows manually reviewed.

Source file	Lines	Responsibility
backend/security/__init__.py	31	Password hashing using the standard library's memory-hard scrypt.
backend/services/__init__.py	612	Authorized application services. Session state is never the data store.
backend/services/care.py	238	Human-authored daily context, personal priorities and clinician follow-up.
backend/services/pilot.py	537	Authorized clinic feasibility pilots, never invented clinical evidence.
backend/services/privacy.py	106	Explicit monitoring withdrawal and visible human-managed data requests.
backend/services/reporting.py	146	Authorized, idempotent patient-report persistence using the caller's session.
backend/services/validation.py	243	Owner-controlled validation protocol and immutable source measurements.
backend/storage/__init__.py	83	Bounded private uploads. Client filenames never become storage paths.
ml/__init__.py	1	Reproducible, non-diagnostic measurement services for ParkinSense-PK.
ml/audio/__init__.py	3	Package exports and import namespace.
ml/audio/features.py	169	Local WAV quality checks and reproducible acoustic measurements.
ml/common/__init__.py	1	Shared result validation.

Exact definitions, line numbers and SHA-256 for each file are retained in component-inventory.json. Package __init__ files also appear so the source inventory is explicit.

Appendix A / Python component map

Files 49-60 of 90. Paths relative to project root. All inventoried via AST; critical workflows manually reviewed.

Source file	Lines	Responsibility
ml/common/guidance.py	491	Local, source-aware patient next steps for actual analysis outcomes.
ml/common/guidance_urdu.py	213	Fixed native Urdu product guidance; no automatic translation or diagnosis.
ml/common/results.py	34	Exports/helpers: numeric_features, normalize_language
ml/common/statistics.py	293	Display definitions and descriptive statistics of accepted module measurements.
ml/common/tables.py	36	Aligned, quality-gated developer feature tables without raw patient content.
ml/inference/__init__.py	3	Package exports and import namespace.
ml/inference/evidence.py	47	Opt-in, versioned approved-document retrieval. Empty by default.
ml/inference/models.py	57	Trusted offline baseline artifact inference and explicit advanced availability.
ml/inference/patient_report.py	308	Patient analysis summaries made from saved measurements and explicit sources.
ml/inference/reports.py	73	Deterministic auditable quality-to-report coordination; no external LLM.
ml/multimodal/__init__.py	3	Package exports and import namespace.
ml/multimodal/fusion.py	35	Exports/helpers: modality_for, fuse_results

Exact definitions, line numbers and SHA-256 for each file are retained in component-inventory.json. Package __init__ files also appear so the source inventory is explicit.

Appendix A / Python component map

Files 61-72 of 90. Paths relative to project root. All inventoried via AST; critical workflows manually reviewed.

Source file	Lines	Responsibility
ml/nlp/__init__.py	3	Package exports and import namespace.
ml/nlp/functional.py	91	Transparent multilingual self-report extraction; no unconfigured ASR/LLM.
ml/tabular/__init__.py	3	Package exports and import namespace.
ml/tabular/adl.py	37	Exports/helpers: analyze_adl
ml/tabular/training.py	175	Small-data baseline training with patient-disjoint holdouts and provenance.
ml/temporal/__init__.py	3	Package exports and import namespace.
ml/temporal/baseline.py	67	Compare aligned measurements against the first compatible accepted result.
ml/validation/__init__.py	1	Measurement validation, separate from disease prediction and clinical scores.
ml/validation/evaluation.py	175	Participant-balanced measurement accuracy and matched repeatability.
ml/vision/__init__.py	2	Task-specific, CPU landmark measurements for functional monitoring.
ml/vision/features.py	260	Conservative motion proxies; no clinical score or disease prediction.
ml/vision/live_overlay.py	201	Focused live capture guides anchored to observed image landmarks.

Exact definitions, line numbers and SHA-256 for each file are retained in component-inventory.json. Package __init__ files also appear so the source inventory is explicit.

Appendix A / Python component map

Files 73-84 of 90. Paths relative to project root. All inventoried via AST; critical workflows manually reviewed.

Source file	Lines	Responsibility
ml/vision/pipeline.py	581	Bounded, streaming video analysis shared by uploaded and live recordings.
ml/vision/spatial.py	153	Truthful, bounded model-local 3D geometry for display only.
ml/vision/tasks.py	17	Recording instructions and minimum visibility for each independent task.
ml/xai/__init__.py	3	Package exports and import namespace.
ml/xai/explain.py	27	Exports/helpers: explain_measurements
scripts/__init__.py	1	Local setup and validation utilities.
scripts/account_recovery.py	42	Issue a single-use recovery code after a clinic has verified the account owner.
scripts/backup_private.py	243	Encrypted database + private-media snapshot, requiring a paused-write maintenance window.
scripts/bootstrap_admin.py	28	Create the first administrator interactively; never ship default credentials.
scripts/check_public_recordings.py	49	Re-run installed, attributed demos without touching patient records.
scripts/download_landmark_models.py	69	Explicitly download the small official MediaPipe landmark model bundles.
scripts/evaluate_recordings.py	85	Evaluate an explicitly supplied, deidentified recording manifest offline.

Exact definitions, line numbers and SHA-256 for each file are retained in component-inventory.json. Package __init__ files also appear so the source inventory is explicit.

Appendix A / Python component map

Files 85-90 of 90. Paths relative to project root. All inventoried via AST; critical workflows manually reviewed.

Source file	Lines	Responsibility
scripts/export_features.py	56	Authorized developer export into ignored local feature storage.
scripts/restore_drill.py	171	Authenticate and restore a snapshot into a NEW isolated SQLite/media directory only.
scripts/run_mobile_api.py	23	Start the native Android gateway; the existing web server is independent.
scripts/secure_media_server.py	190	Small loopback media gateway. Put behind a trusted TLS reverse proxy in clinics.
scripts/setup_local.py	28	Generate an ignored local PostgreSQL configuration without printing secrets.
scripts/train_baseline.py	37	Offline developer CLI; supply an approved table, never patient raw media.

Exact definitions, line numbers and SHA-256 for each file are retained in component-inventory.json. Package __init__ files also appear so the source inventory is explicit.

Appendix B / Native and runtime assets

Path / component	Responsibility
mobile/android/.../MainActivity.java	Patient UI, account flow, dashboard/history/report and forms.
mobile/android/.../LiveCheckerActivity.java	CameraX / AudioRecord lifecycle, temporary capture, quality preview and upload choice.
mobile/android/.../ApiClient.java	Bounded authenticated requests, no redirects, origin binding and callback invalidation.
mobile/android/.../TokenStore.java	Keystore AES-GCM token persistence; server change clears auth.
mobile/android/.../ServerAddress.java	HTTPS/public and private-LAN origin validation.
mobile/android/.../SignalMath.java	Measured local audio/image quality math, not landmark or disease analysis.
mobile/android/.../DarkUi.java	Shared native styling and controls.
app/components/live_plot.js	Persistent Plotly host, queued updates, interaction and teardown lifecycle.
AndroidManifest.xml + res/xml/	Permissions, exported activities, disabled backups and release/debug network policy.
Gradle files + wrapper	Pinned build toolchain, SDK/CameraX dependencies and wrapper integrity.
infra/compose.yaml	PostgreSQL 16 service, loopback binding, persistent volume and health check.
.streamlit/config.toml	Dark theme, 100 MB upload, XSRF protection, disabled usage telemetry.

Android Java paths abbreviated above all resolve under mobile/android/app/src/main/java/org/parkinsense/pk/. Debug network override lives under app/src/debug/res/xml/.

Appendix C / Setup, operational and test assets

Asset	Role / audit treatment
pyproject.toml; uv.lock; requirements.txt	Python runtime/development dependencies; reproducible locked environment, Python 3.11.
scripts/run_local.sh	Starts existing database, media gateway and web interface; cleans up child gateway on exit.
scripts/build_android_apk.sh	Build/test/lint, APK copy, signature verification and SHA-256.
scripts/setup_local.py / bootstrap_admin.py	Local configuration and first admin initialization; no shipped shared password.
scripts/download_landmark_models.py	Official MediaPipe bundles and local model installation.
tests/*.py	30 test modules listed in automated inventory, including new baseline UI regression file.
Android src/test/...	ServerAddressTest and SignalMathTest; 12 test cases.
docs/ and README.md	Usage and implementation notes; older historical test totals are superseded by this audit.
data/private; models/landmarks; .env	Private/runtime paths excluded from report contents and ordinary source inventory.
artifacts/project-audit-2026-10-18/	This audit's local test logs, metadata, results and helper scripts.
output/pdf/	Final detailed report; layout rendered and visually inspected before delivery.

Final assessment

Project mein six assessment sections, local measurement pipelines, estimated 3D review, descriptive statistics, saved reports aur authorized care workflow implemented aur software-tested hain. Ab sab se valuable investment matching-task clinical evidence, patient usability aur operational validation hai. Is report ko engineering baseline aur clinic discussion guide ke taur par use karein.